📄 HADOOP & HIVE DATA ANALYSIS REPORT

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1. Cover Page

Title: Hadoop & Hive Data Analysis
Subject: Big Data Analytics

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2. Problem Statement

The objective is to analyze employee data stored in HDFS to understand cost-to-company and country-wise employee distribution. The dataset contains the fields: **empId, employeeAge, salary, country**.

The task involves:

- Performing file operations using Hadoop HDFS
- Creating and loading a Hive table
- Analyzing employee count country-wise

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3. Task 1: HDFS Operations

i) Create Directory in HDFS

hdfs dfs -mkdir /PartBTest

Description: Created a directory named **PartBTest** in HDFS.

```
PS C:\Users\HP> docker cp "C:\Users\HP\Downloads\Module_4_Assignment_Question\Module 4 Assignment Question\problem-1-dat
a.txt" a225774264ec:/root/
Successfully copied 3.58kB to a225774264ec:/root/
PS C:\Users\HP> docker exec -it a225774264ec /bin/bash
root@a225774264ec:/# ls /root
problem-1-data.txt
```

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ii) Upload File to HDFS

hdfs dfs -put problem-1-data /PartBTest/

Description: Uploaded the dataset from local system to HDFS.

```
root@a225774264ec:/# hadoop fs -put /root/problem-1-data.txt /PartBTest
2026-04-03 13:50:21,641 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteH
ostTrusted = false
root@a225774264ec:/# hadoop fs -ls /PartBTest
Found 1 items
-rw-r--r--   3 root supergroup       1717 2026-04-03 13:50 /PartBTest/problem-1-data.txt
```

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iii) List Files in Directory

hdfs dfs -ls /PartBTest

Description: Displayed all files present in the directory.

```
root@a225774264ec:/# hadoop fs -ls /PartBTest
Found 1 items
-rw-r--r--   3 root supergroup       1717 2026-04-03 13:50 /PartBTest/problem-1-data.txt
```

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iv) Copy File to Another Directory

hdfs dfs -mkdir /PartBTest2
hdfs dfs -cp /PartBTest/problem-1-data /PartBTest2/

Description: Copied the file into another directory.

```
root@a225774264ec:/# hadoop fs -mkdir /PartBTest2
mkdir: `/PartBTest2': File exists
root@a225774264ec:/# hadoop fs -cp /PartBTest/problem-1-data.txt /PartBTest2
2026-04-03 13:51:23,110 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteH
ostTrusted = false
2026-04-03 13:51:23,252 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteH
ostTrusted = false
root@a225774264ec:/# hadoop fs -ls /PartBTest2
Found 1 items
-rw-r--r--   3 root supergroup       1717 2026-04-03 13:51 /PartBTest2/problem-1-data.txt
```

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4. Task 2: Hive Operations

i) Create Hive Table

```
CREATE TABLE employee_data (
   empId INT,
   employeeAge INT,
   salary FLOAT,
   country STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE;
```

Description: Created a structured Hive table to store employee data.

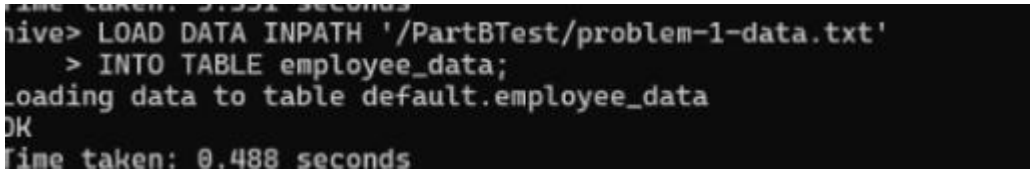
```
Hive-on-MR is deprecated in Hive 2 and may not be availab
hive> CREATE TABLE employee_data (
    > empId INT,
    > employeeAge INT,
    > salary INT,
    > country STRING
    > )
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY ',';
OK
Time taken: 3.967 seconds
```

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ii) Load Data into Hive Table

```
LOAD DATA INPATH '/PartBTest/problem-1-data'
INTO TABLE employee_data;
```

Description: Loaded data from HDFS into Hive table.

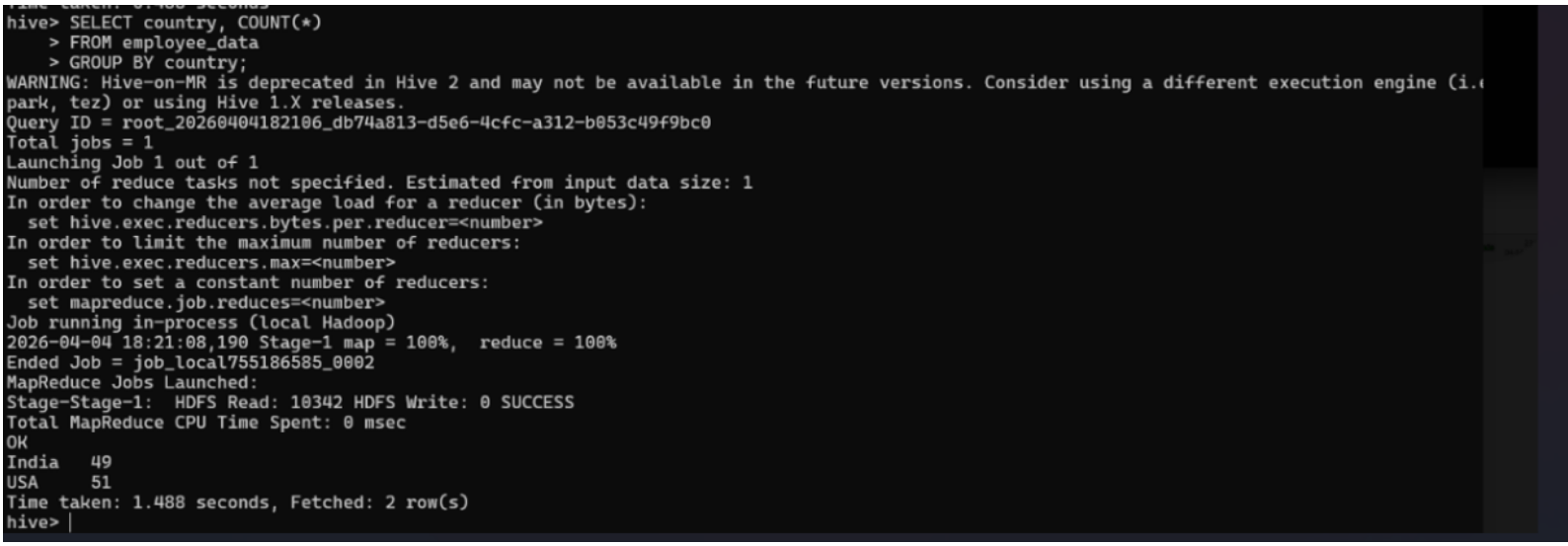


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iii) Country-wise Employee Count

SELECT country, COUNT(*) AS total_employees
FROM employee_data
GROUP BY country;

Description: Calculated number of employees for each country.



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5. Output Explanation

- The query groups employee records based on country
- It counts total employees in each country
- Helps the company analyze workforce distribution
- Useful for making expansion and business decisions

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6. Conclusion

- Successfully performed HDFS operations like directory creation, upload, and copy
- Created and managed Hive tables
- Loaded data from HDFS into Hive
- Performed analytical query using HQL
- Understood how Big Data tools help in real-world business analysis